

Data Engineer / Senior Backend Engineer

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 Tokyo, Japan

Visa: Engineer / Specialist in Humanities / International Services (3-year term)

Showcase: [Olist Lakehouse Pipeline](#) | [perapera.me](#) | [PeraTube](#)

Professional Summary

Data Engineer & Senior Backend Engineer with **6+ years** of specialized experience in large-scale distributed systems, high-concurrency database migrations, and cloud-native architecture, backed by a 20-year software engineering foundation. Actively transitioning into modern **Lakehouse Data Engineering**, possessing in-depth practical mastery in **Databricks, PySpark, Delta Lake (Medallion Architecture, Lakeflow Pipelines, Unity Catalog, CDC/SCD Type 2)**. Proven track record in orchestrating multi-layer data pipelines, dimensional modeling, and closing the loop between enterprise data lakes and real-time operational applications. Highly proficient in the Spring ecosystem, NoSQL (Cassandra, Couchbase), and AI-augmented engineering workflows.

Technical Skills

Core Competencies

- **Data Engineering & Lakehouse:** **Databricks, PySpark, Delta Lake** (ACID, MERGE INTO, Time Travel, CDF, Liquid Clustering), **Auto Loader, Databricks Asset Bundles (DABs), Lakeflow Pipelines (DLT), Unity Catalog**, Medallion Architecture (Bronze/Silver/Gold), **SCD Type 2, CDC**, Dimensional Modeling (Star Schema), Window Deduplication, SQL
- **Distributed Backend & Data Systems:** Java, Kotlin, Python (FastAPI), **Spring Boot (2.x/3.x), Kafka (CDC), Cassandra, Couchbase, Snowflake, Oracle, Redis, RabbitMQ, Microservices, DDD**
- **Search & Real-Time Sync:** **Meilisearch**, Convex (Reactive WebSocket Push), Elasticsearch/Vector Search, RESTful APIs
- **Cloud & DevOps:** **Azure (Static Web Apps, AKS), Railway**, AWS, Docker, Minikube, Podman, CI/CD (GitHub Actions, GitLab CI/CD)
- **AI-Augmented Engineering:** Proficient in **Claude Code, Codex, and Antigravity** for rapid development, code generation, and pipeline refactoring

Languages & Frameworks

- **Languages:** Python, Java, Kotlin, SQL, TypeScript, C#, Go, Rust, C++
 - **Web & Frontend:** Next.js, React.js, Vue.js, Convex, Tailwind CSS
 - **Languages (Human):** **Japanese (Business), English (Business)**, Chinese (Native)
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Selected Projects

Olist Lakehouse Pipeline — Production-Grade Incremental Lakehouse on Databricks

Data Engineering Showcase | Architect & Data Engineer

Repository: <https://github.com/kidwon/olist-lakehouse-pipeline>

- **Production-Grade Medallion Architecture:** Built an incremental Medallion Lakehouse on Databricks processing 100k+ real Brazilian e-commerce orders (Olist dataset), replaying daily feeds from Landing through Bronze, Silver, Gold, and Business Marts.
- **Idempotent Ingestion & CDC Handling:**
 - Leveraged **Auto Loader** with checkpointing and `_rescued_data` for schema evolution and zero data loss on unexpected fields (`discount_amount`).
 - Implemented **forward-only CDC MERGE** using sequence tracking (`change_seq`) to ensure out-of-order state updates never regress order statuses.
 - Eliminated within-batch and cross-batch re-delivery duplicates via window deduplication (`row_number`) and INSERT-only MERGE.
- **Hand-Written SCD Type 2 & Point-in-Time Joins:** Designed a change-hash detection mechanism to close historical records and insert new versions in a single atomic MERGE for seller location tracking, joining fact tables (`fact_order_item`) by transaction date to prevent metric drift.
- **Data Quality Gates & Automated Quarantine:** Established per-rule quarantine tables (`ops.quarantine`) and an automated execution gate (`ops.dq_gate`) that halts downstream Gold publication if anomaly rates exceed 5%, logging comprehensive operational metrics (`ops.dq_metrics`).
- **Star Schema Modeling & Business Mart:** Structured Gold into a Kimball Star Schema (`fact_order_item`, `dim_date`, `dim_customer` resolved via `customer_unique_id`, `dim_product`, `dim_seller`) and built seller delivery performance marts (GMV, on-time delivery rate, rating trends).
- **CI/CD, Testing & Decision Records:** Packaged the 7-task serverless workflow via **Databricks Asset Bundles (DABs)**; implemented 28 automated unit and end-to-end tests (`pytest`) validating 100% anomaly injection/detection parity; documented 8 comprehensive Architectural Decision Records (ADRs).

perapera.me — High-Performance Japanese Learning Ecosystem

Personal Project | Lead Full-Stack Architect

- **Architecture:** Built a hybrid multi-service platform using **Next.js/Convex** and **FastAPI**.
- **Search Engine:** Integrated **Meilisearch** for high-performance search across millions of records.
- **Resilient AI:** Implemented a multi-tier fallback architecture (Cache/Key Pools/DeepSeek) to handle strict LLM rate limits.
- **Cloud & Payments:** Orchestrated CI/CD pipelines via GitHub Actions to **Azure** and **Railway**, and integrated **Stripe** for subscriptions.

PeraTube — Japanese Video Example-Sentence Search API (Developer SaaS)

Personal Project | Solo Full-Stack Architect

- **Product:** Indexes massive YouTube subtitle corpora into searchable segments — developers query how Japanese is actually used in real videos via API, receiving video ID, timestamp, transcript, and surrounding context with millisecond latency.

- **Architecture:** **Next.js** dashboard/playground + **Python FastAPI** orchestration layer + **Convex** as the single source of truth (users, quotas, API keys) + **Meilisearch** full-text search.
- **Commercialization:** **Stripe** subscriptions and credit packs, API-key authentication (sha256-hashed, LRU-cached validation), atomic quota deduction in Convex, admin dashboard, and content moderation.
- **Data Pipeline & Quality:** YouTube → VTT subtitles → segment parsing → chunked **.jsonl** → batch import; 15+ pytest test files across the API and pipeline; deployed on **Railway** with a Cloudflare custom domain.

Jessie Signal — Real-Time Market Sentiment Dashboard

Personal Project | Full-Stack Architect

- Built a **React / Convex** serverless web app that aggregates live market data (**VIX, CNN Fear & Greed, major ETFs**) and classifies the market into actionable sentiment scenarios.
- Implemented **user authentication (Clerk)** so users can log in and **subscribe to email alerts** on market-scenario changes, delivered via **Resend**.
- Added a **server-side caching** layer for instant page loads and a gated **admin dashboard**; deployed on **Railway**.

Professional Experience

Contract Backend Engineer | NTT DATA

April 2025 - August 2026 | Tokyo, Japan

Insurance System Modernization Project (Struts to Spring Migration)

Responsibilities:

- Contributed to the full-stack modernization of a core insurance application across two major phases, transitioning from legacy frameworks to a modern web application stack:
 - **Legacy Source Stack:** TERASOLUNA 2.2.0 (Struts 1.x) + Oracle 11g + intra-mart 7.2 + JDK 1.7
 - **Target Modernized Stack:** TERASOLUNA 5.x (Spring MVC) + Oracle 18c XE + intra-mart 8.0.36 + JDK 17 (*Architecture Migration, Framework Upgrade, Spring Ecosystem*)
- **Phase 1 (UI/UX Optimization):** Refactored legacy **JSP** templates and adjusted intra-mart custom tags to improve front-end usability, while maintaining existing Struts backend filters and Java logic. (*JSP Refactoring, intra-mart Ecosystem, Struts 1.x*)
- **Phase 2 (Architectural Upgrade):** Refactored backend business logic layers into clean, decoupled, and testable Spring-managed beans using **Dependency Injection** and modern Spring paradigms. (*Spring Core, Java 17, Backend Engineering*)

Technologies & Environment: Java 17, Spring MVC (TERASOLUNA 5.x), Struts 1.x (TERASOLUNA 2.2.0), intra-mart 7.2 / 8.0.36, JSP, Oracle 11g / 18c XE, Eclipse.

Contract Backend Engineer | Rakuten Group, Inc.

September 2021 - March 2025 | Tokyo, Japan

Membership System & Database Migration (Kotlin/Spring Boot Microservices)

Responsibilities:

- Conducted a critical Proof of Concept (PoC) to test if Couchbase could handle high-concurrency and high-traffic conditions, generating 300 million mock membership records for benchmarking. *(PoC, Database Benchmarking, High-Concurrency Testing)*
- Participated in executing the database migration from Couchbase to **Cassandra** after deep stress-testing revealed Couchbase's performance degradation and high error rates during node rebalancing under concurrent load. *(Database Migration, NoSQL, Fault-Tolerance)*
- Developed a high-performance benchmarking API server using the asynchronous **Vert.x** framework to generate and process the 300 million mock records for load testing. *(Vert.x, Data Generation, Backend Development)*
- Executed distributed stress testing in the CaaS environment using **Gatling**, integrating automated test scenarios into **GitLab CI/CD** pipelines. *(Gatling, Load Testing, GitLab CI/CD)*
- Developed and packaged a custom **Argon2** hashing library to standardize secure data operations across multiple microservices. *(Kotlin, Argon2, Security)*
- Contributed to code refactoring across membership services, moving duplicate data-access logic into shared core modules to reduce technical debt. *(Code Refactoring, Clean Code)*
- Designed and automated integration and regression test suites for the migrated RESTful endpoints using **Postman** and **Newman** across local **MiniKube** and staging environments. *(Postman, Newman, API Testing, MiniKube)*
- Updated **OpenAPI (Swagger)** documentation to ensure clear API specifications for cross-team collaboration during the database migration. *(OpenAPI, Swagger, API Documentation)*

Technologies & Environment: Kotlin, Java, Spring Boot, Vert.x, Cassandra, Couchbase, MariaDB, Gatling, Postman, Newman, GitLab CI/CD, Podman, MiniKube, IntelliJ IDEA.

E-commerce Platform (Rakuten Fashion - Agency Order Management System)

Responsibilities:

- Joined the project in its latter phase to contribute to the development and optimization of the Rakuten Fashion Agency Order Management System. *(E-commerce Operations, Order Pipelines)*
- Developed lightweight **Spring Boot scheduled batch tasks** using **@Scheduled** and **RabbitMQ** to automate error handling, periodically routing stalled or unprocessed orders back into the message queue for asynchronous re-processing. *(Spring Boot, @Scheduled, RabbitMQ, Asynchronous Processing)*
- Audited and streamlined legacy **Gradle** build configurations and internal API version management across multiple microservice repositories utilizing **JFrog Artifactory**, resolving version dependency conflicts. *(Gradle, JFrog Artifactory, Dependency Management)*
- Standardized API specifications by integrating the native **Spring Boot OpenAPI (Swagger)** extension across all backend services, establishing a clear contract for cross-team collaboration. *(Spring Boot, OpenAPI, Swagger, API Standardization)*
- Conducted initial business domain analysis by updating enterprise technical specifications in Confluence and maintaining relational database schemas and **DDL** migration documentation.

(Technical Documentation, Confluence, DDL, Database Schema Management)

Technologies & Environment: Java, Spring Boot, RabbitMQ, Gradle, JFrog Artifactory, OpenAPI, Swagger, MariaDB, Confluence.

Earlier Experience

- **Full-time Senior Software Engineer | Techoes Inc., Tokyo** (Jan 2020 - Aug 2021)
 - Participated in backend development for a multiplayer game, leading the online chat system and AI pathfinding modules using **Go, gRPC, and Protobuf** for high-concurrency, low-latency microservices.
 - Contributed to a live streaming platform, transitioning from interactive front-end development (**Vue.js, Pixi.js**) to backend API development and performance optimization (**PHP/Laravel**).
- **Contract Software Engineer | NEC, Tokyo** (Aug 2018 - Oct 2019)
 - Developed VMS (Video Management System) integrations for the National Police Agency, engineering C# module extensions on top of the **Milestone XProtect VMS** enterprise platform (Windows/VS2017).
 - Engineered backend APIs, web interfaces, and scheduled batch tasks for the **Mobile Suica system**, taking full ownership from core implementation to designing unit test specifications and executing test suites.
- **Lead Client Engineer | Shanghai Thinky Game, China** (Jul 2016 – Feb 2018)
 - Architected and led client-side development for 3D mobile tank battle game **坦克雄心** (**Unity 5.x, C#**).
 - Engineered core combat systems including dynamic shell trajectory physics, armor damage penetration, hit-detection, visual FX managers, dynamic weather systems, and custom editor tools.
 - Conducted deep low-level profiling and optimization across Memory, CPU, and GPU pipelines to achieve high frame rates on low-spec devices.
- **Client Engineer | Shanghai Haymaker Game, China** (Dec 2015 – Jun 2016)
 - Developed core battle logic and character skill execution systems for mobile action RPG **小李飞刀** (**Unity 4.x, C#, SVN**), focusing on high-frequency action inputs and state management.
- **UI & Client Engineer | 2K Games China, China** (Mar 2015 – Nov 2015)
 - Engineered event-driven UI frameworks and HUD systems for AAA PC title *Borderlands Online* (**Unity 4.x, ActionScript 3, C#**), integrating Scaleform/Flash interfaces with C++ low-level game engines using **Perforce**.
- **Client Engineer | Shanghai Hanniu Game, China** (Mar 2014 – Feb 2015)
 - Built real-time combat modules and skill UI systems for mobile MOBA title **暴走DOTA** (**Unity 4.x, C#**), optimizing client-side network packet serialization and rendering performance.

- **UI & Client Engineer | Limbo Studio, China** (Apr 2011 – Mar 2014)
 - Developed PC poker/card gaming platform 新安游 UI and client logic using **C++**, **ActionScript 2**, and **Flash**.
 - Engineered client UI modules and HUD mechanics for PC TPS title 枪神 using **Unreal Engine 2.5 (C++, ActionScript 3)**, interfacing UI components with native UE C++ core modules.
 - **UI Programmer | The9 Limited, China** (Dec 2007 – Mar 2011)
 - Developed interactive client UI modules and business logic for PC casual online title 小小部落, leveraging **Gamebryo C++ engine**, **ActionScript 2**, and **Flash** pipelines.
 - **Game QA & Operations Specialist | Wemade Shanghai, China** (Oct 2003 – Dec 2007)
 - Managed Game Master (GM) operations and technical QA for PC MMORPG *Legend of Mir 3* (传奇3), conducting server load verification, live test server environment troubleshooting, and operational feedback loops.
 - **Computer Science Teacher | Tiantai Chengxi High School, China** (Sep 2002 – Sep 2003)
 - Taught computer science fundamentals and basic software usage, developing strong technical communication and presentation skills.
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Education

- **Zhejiang University of Technology**
 - Computer Application and Maintenance